

# Slide 1: Introduction to Python Arrays

- An array stores multiple values in one variable.
- All elements usually have the same data type.
- In Python, we commonly use lists as arrays.
- Example: `numbers = [10, 20, 30, 40]`

## Slide 2: Creating an Array

- Using List: `arr = [1, 2, 3, 4, 5]`
- Using array module:
- `import array`
- `arr = array.array('i', [1, 2, 3, 4])`
- 'i' means integer type.

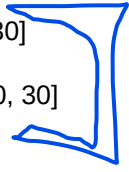


## Slide 3: Accessing Elements

- Index starts from 0.
- Example:
- `arr = [10, 20, 30, 40]`
- `print(arr[0])` # 10
- `print(arr[2])` # 30

## Slide 4: Updating Elements

- `arr = [10, 20, 30]`
- `arr[1] = 50`
- Output: `[10, 50, 30]`



## Slide 5: Adding Elements

- `append()` → Add at end: `arr.append(60)`
- `insert()` → Add at specific position: `arr.insert(1, 25)`

→  $\begin{matrix} 0 & 1 & 2 \\ [0, & 2, & 4] \end{matrix}$  ←

$[0, 2, 4, 60]$

$[0, 25, 2, 4]$

$[0, 25, 2, 4]$

## Slide 6: Removing Elements

- `remove()` → Remove by value: `arr.remove(20)`
- `pop()` → Remove by index: `arr.pop(1)`

$[0, 20, 1, 5]$   $[0, 1, 5]$   
1 1 1 1  
[0, 1, 5]

## Slide 7: Array Length

- Use len(arr) to get total number of elements.

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## Slide 8: Looping Through Array

- for item in arr:
- print(item)

for i, item in arr:



## Slide 9: Basic Array Operations

- sum(arr) → Sum of elements
- max(arr) → Maximum value
- min(arr) → Minimum value

## Slide 10: Sorting Array

- `arr.sort()` → Sort in ascending order
- `arr.sort(reverse=True)` → Sort in descending order

`[1, 0, 9, 2]`

`[0, 1, 2, 9]`

5 2 1 0